PREMIER UNIVERSITY, CHITTAGONG

Department of Computer Science and Engineering

LAB REPORT

BoardGameGeek Data Analysis on Hadoop

Using Hadoop MapReduce and Apache Pig

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1 Problem Statement

The objective of this laboratory exercise is to process and analyze a BoardGameGeek dataset using Apache Hadoop, HDFS, MapReduce and Apache Pig. The selected dataset contains information such as game ID, title, category, publisher, rating count, ownership count and comments.

In total, three MapReduce programs and five Pig analyses were executed. The main tasks were to identify the most-owned game, summarize rating and engagement information, count games by publisher, and perform global and category-wise ranking using Apache Pig.

2 Dataset Description

The dataset used in this experiment is based on the `games_detailed_info.csv` file from the BoardGameGeek Reviews dataset on Kaggle.

The raw file contained:

- 21,631 records
- 56 columns
- 0 duplicate game IDs
- 0 malformed CSV rows

For analysis, the dataset was cleaned and reduced to 13 fields:

No.	Attribute	Description
1	gameID	Unique game identifier
2	name	Game title
3	yearPublished	Publication year
4	primaryPublisher	First listed publisher
5	primaryCategory	First listed category
6	minPlayers	Minimum players
7	maxPlayers	Maximum players
8	playingTime	Playing time
9	usersRated	Number of users who rated
10	averageRating	Average rating
11	owned	Ownership count
12	numComments	Number of comments
13	averageWeight	Average complexity value

The final cleaned dataset contained **21,347 records**, **81 primary categories**, and **3,824 primary publishers**.

The category and publisher columns in the original file contain multiple values. To keep one record per game, the first listed category and first listed publisher were used as `primaryCategory` and `primaryPublisher`.

3 Experimental Setup

The experiment was performed on macOS using Hadoop 3.4.1 in pseudo-distributed mode, Apache Pig 0.17.0 and OpenJDK 11.0.32. The cleaned file was uploaded to:

```
/boardgame/boardgames.csv
```

```
Last login: Fri Sep  4 19:53:31 on ttys000
(base) home@Davids-MacBook-Air ~ % jps
hadoop version | head -3
pig -version
hdfs dfs -ls -h /boardgame
20098 DataNode
20242 SecondaryNameNode
20867 NameNode
35619 Jps
20438 ResourceManager
4394 JobHistoryServer
20541 NodeManager
Hadoop 3.4.1
Source code repository https://github.com/apache/hadoop.git -r 4d7825309348956336b8f06a08322b7842284
9b1
Compiled by mthakur on 2024-10-09T14:57Z
Apache Pig version 0.17.0 (r1797386)
compiled Jun 02 2017, 15:41:58
2026-09-04 19:54:09,179 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your pl
atform... using builtin-java classes where applicable
Found 9 items
-rw-r--r--   1 home supergroup      1.8 M 2026-09-04 19:28 /boardgame/boardgames.csv
drwxr-xr-x   - home supergroup          0 2026-09-04 19:31 /boardgame/out_mostowned
drwxr-xr-x   - home supergroup          0 2026-09-04 19:36 /boardgame/out_publisher
drwxr-xr-x   - home supergroup          0 2026-09-04 19:34 /boardgame/out_rating_summary
drwxr-xr-x   - home supergroup          0 2026-09-04 19:42 /boardgame/pig_top10_most_owned
drwxr-xr-x   - home supergroup          0 2026-09-04 19:40 /boardgame/pig_top10_most_rated
drwxr-xr-x   - home supergroup          0 2026-09-04 19:45 /boardgame/pig_top10_owned_by_category
drwxr-xr-x   - home supergroup          0 2026-09-04 19:43 /boardgame/pig_top10_rated_by_category
drwxr-xr-x   - home supergroup          0 2026-09-04 19:38 /boardgame/pig_top5_categories
```

Figure 1: Hadoop environment, running services and HDFS project files

4 MapReduce Analysis

4.1 MR-1: Most Owned Board Game

The purpose of this program was to find the game with the highest ownership count. The mapper reads the game ID, title and `owned` value. A common key is used so that the reducer can compare all candidates and keep the largest value.

The result was:

```
30549    Pandemic    168364
```

Therefore, **Pandemic** was the most-owned game with **168,364 owners**. The same result was independently checked from the cleaned CSV using `awk`.

```
(base) home@Davids-MacBook-Air ~ % echo "MR-1: Most Owned Board Game"
cat ~/boardgame_lab/results/mapreduce/mr1_most_owned.txt
echo
echo "Independent verification:"
awk -F',' 'BEGIN{m=-1} $11>m{m=$11; row=$1"|"$2"|"$5"|"$11} END{print row}' \
~/boardgame_lab/clean/boardgames.csv
MR-1: Most Owned Board Game
30549   Pandemic        168364

Independent verification:
30549|Pandemic|Medical|168364
```

Figure 2: MR-1 most-owned game and independent verification

4.2 MR-2: Rating and Engagement Summary

This program produces one summary record for each game ID. The mapper reads the average rating, number of users who rated the game, and number of comments. The reducer writes the summary for each game.

The output contained exactly **21,347 records**, matching the cleaned dataset.

For Pandemic, the output was:

```
30549    7.58896    109006    17305
```

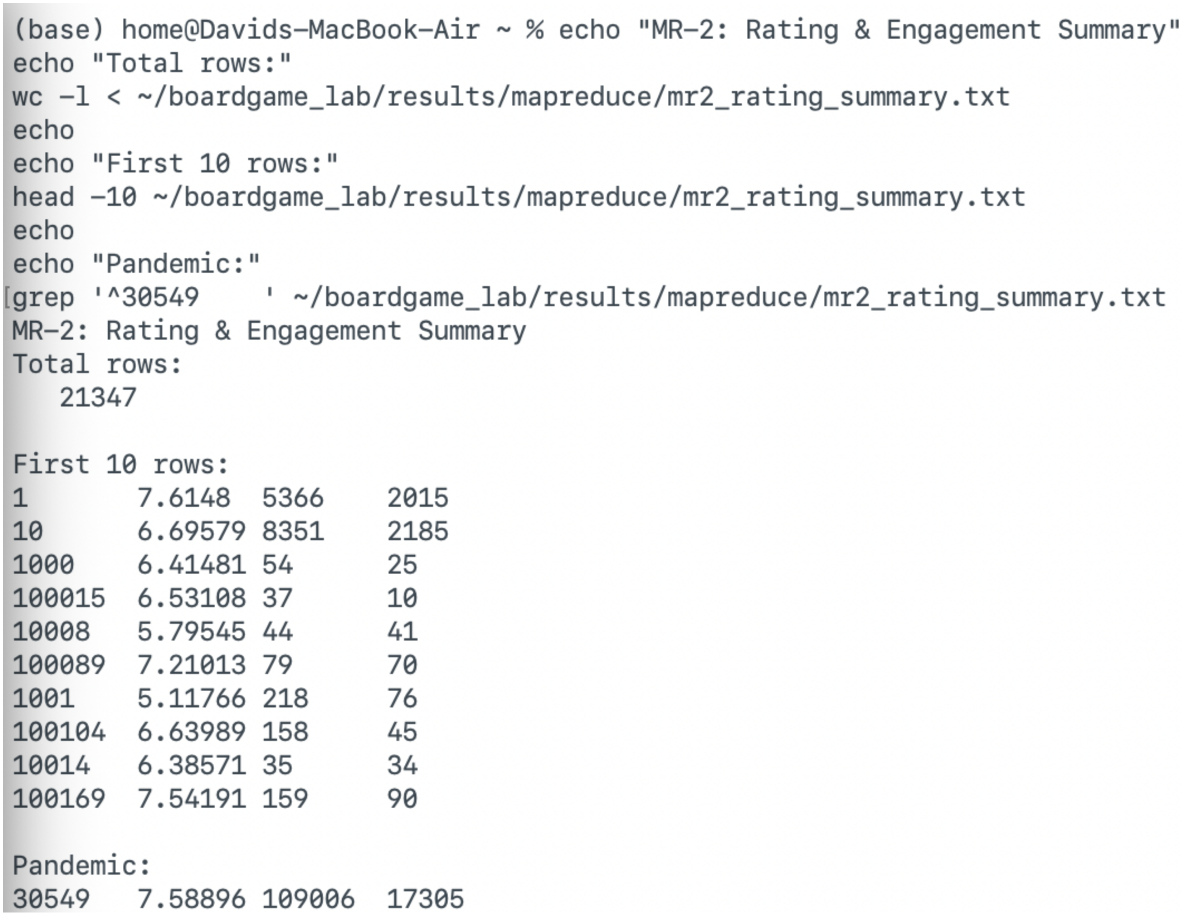

Figure 3: MR-2 rating and engagement summary

4.3 MR-3: Games per Publisher

This MapReduce program counts the number of games associated with each primary publisher. The mapper emits:

```
(primaryPublisher, 1)
```

The reducer adds the values for each publisher. The complete output contained **3,824 primary publishers**.

The top publishers were:

Publisher	Games
(Self-Published)	543
(Web published)	380
Hasbro	379
Decision Games (I)	272
GMT Games	268
Ravensburger	255
AMIGO	249
(Public Domain)	214
KOSMOS	212
999 Games	208

```
(base) home@Davids-MacBook-Air ~ % echo "MR-3: Top 10 Publishers by Number of Games"
sort -t$'\t' -k2 -nr \
~/boardgame_lab/results/mapreduce/mr3_games_per_publisher.txt \
| head -10
echo
echo "Total publishers:"
wc -l < ~/boardgame_lab/results/mapreduce/mr3_games_per_publisher.txt
MR-3: Top 10 Publishers by Number of Games
(Self-Published)        543
(Web published) 380
Hasbro  379
Decision Games (I)      272
GMT Games       268
Ravensburger    255
AMIGO   249
(Public Domain) 214
KOSMOS  212
999 Games       208

Total publishers:
    3824
```

Figure 4: MR-3 top publishers and total publisher count

5 Apache Pig Analysis

Apache Pig was used for ranking, grouping and category-based analysis. During testing, Pig 0.17.0 had a compatibility problem with a top-level `ORDER BY` operation on Hadoop 3.4.1. The working scripts therefore perform sorting inside a nested `FOREACH` block after grouping the data.

5.1 Pig-1: Top 5 Primary Categories

This analysis groups games by primary category, counts the games in each category and returns the five largest categories.

The result was:

Category	Games
Card Game	4615
Abstract Strategy	1544
Adventure	1144
Action / Dexterity	1057
Animals	1049

Card Game was therefore the most frequent primary category.

```
(base) home@Davids-MacBook-Air ~ % echo "Pig-1: Top 5 Primary Categories"
cat ~/boardgame_lab/results/pig/pig1_top5_categories.txt
Pig-1: Top 5 Primary Categories
Card Game|4615
Abstract Strategy|1544
Adventure|1144
Action / Dexterity|1057
Animals|1049
```

Figure 5: Pig-1 top five primary categories

5.2 Pig-2: Top 10 Most-Rated Games

This Pig program sorts games according to `usersRated` and selects the top ten. Here, "most-rated" means the highest number of submitted ratings, not the highest average score.

The first result was:

```
30549 | Pandemic | Medical | 109006
```

Pandemic therefore had the largest number of user ratings in the cleaned dataset.

```
(base) home@Davids-MacBook-Air ~ % echo "Pig-2: Top 10 Most-Rated Games"
cat ~/boardgame_lab/results/pig/pig2_top10_most_rated.txt
Pig-2: Top 10 Most-Rated Games
30549|Pandemic|Medical|109006
822|Carcassonne|City Building|108776
13|Catan|Economic|108064
68448|7 Wonders|Ancient|90021
36218|Dominion|Card Game|81582
9209|Ticket to Ride|Trains|76207
178900|Codenames|Card Game|74456
167791|Terraforming Mars|Economic|74269
173346|7 Wonders Duel|Ancient|69528
31260|Agricola|Animals|66107
```

Figure 6: Pig-2 top ten most-rated games

5.3 Pig-3: Top 10 Most-Owned Games

This analysis sorts all games according to the `owned` field and keeps the first ten records.

The highest result was:

```
30549 | Pandemic | Medical | 168364
```

This matches the result produced by MR-1.

```
(base) home@Davids-MacBook-Air ~ % echo "Pig-3: Top 10 Most-Owned Games"
[cat ~/boardgame_lab/results/pig/pig3_top10_most_owned.txt
Pig-3: Top 10 Most-Owned Games
30549|Pandemic|Medical|168364
13|Catan|Economic|167733
822|Carcassonne|City Building|161299
68448|7 Wonders|Ancient|120466
178900|Codenames|Card Game|119753
173346|7 Wonders Duel|Ancient|111275
36218|Dominion|Card Game|106956
9209|Ticket to Ride|Trains|105748
167791|Terraforming Mars|Economic|101872
129622|Love Letter|Card Game|98395
```

Figure 7: Pig-3 top ten most-owned games

5.4 Pig-4: Top 10 Most-Rated Games by Category

This Pig job first groups games according to `primaryCategory`. Within each category, games are sorted by `usersRated` and the top ten are retained.

The dataset contains **81 primary categories**. The program generated **800 output records**. The total is lower than 810 because some categories contain fewer than ten games.

For the Card Game category, the highest result was:

```
36218 | Dominion | Card Game | 81582
```

```
(base) home@Davids-MacBook-Air ~ % echo "Pig-4: Top 10 Most-Rated Games by Category"
echo "Total output rows:"
wc -l < ~/boardgame_lab/results/pig/pig4_top10_rated_by_category.txt
echo
echo "Card Game:"
awk -F'|' '$3=="Card Game"' \
~/boardgame_lab/results/pig/pig4_top10_rated_by_category.txt
echo
echo "Distinct categories:"
awk -F'|' '{print $3}' \
~/boardgame_lab/results/pig/pig4_top10_rated_by_category.txt \
[| sort -u | wc -l
Pig-4: Top 10 Most-Rated Games by Category
Total output rows:
     800

Card Game:
36218|Dominion|Card Game|81582
178900|Codenames|Card Game|74456
148228|Splendor|Card Game|64734
129622|Love Letter|Card Game|59484
39856|Dixit|Card Game|54525
28143|Race for the Galaxy|Card Game|48919
1927|Munchkin|Card Game|43759
98778|Hanabi|Card Game|41588
11|Bohnanza|Card Game|40113
50|Lost Cities|Card Game|39535

Distinct categories:
      81
```

Figure 8: Pig-4 top most-rated games by category and output count

5.5 Pig-5: Top 10 Most-Owned Games by Category

The final Pig analysis groups games by primary category and finds the ten games with the highest ownership values within each category.

The generated output contained **800 records** across **81 primary categories**.

For the Card Game category, the highest result was:

```
178900 | Codenames | Card Game | 119753
```

```
(base) home@Davids-MacBook-Air ~ % echo "Pig-5: Top 10 Most-Owned Games by Category"
echo "Total output rows:"
wc -l < ~/boardgame_lab/results/pig/pig5_top10_owned_by_category.txt
echo
echo "Card Game:"
awk -F'|' '$3=="Card Game"' \
~/boardgame_lab/results/pig/pig5_top10_owned_by_category.txt
echo
echo "Distinct categories:"
awk -F'|' '{print $3}' \
~/boardgame_lab/results/pig/pig5_top10_owned_by_category.txt \
| sort -u | wc -l
Pig-5: Top 10 Most-Owned Games by Category
Total output rows:
     800

Card Game:
178900|Codenames|Card Game|119753
36218|Dominion|Card Game|106956
129622|Love Letter|Card Game|98395
148228|Splendor|Card Game|92366
1927|Munchkin|Card Game|78849
39856|Dixit|Card Game|76535
98778|Hanabi|Card Game|68643
133473|Sushi Go!|Card Game|65495
11|Bohnanza|Card Game|60282
28143|Race for the Galaxy|Card Game|59758

Distinct categories:
      81
```

Figure 9: Pig-5 top most-owned games by category and output count

6 Summary of Results

Analysis	Final Result
MR-1: Most Owned Game	Pandemic – 168,364 owners
MR-2: Rating Summary	21,347 output records
MR-3: Games per Publisher	(Self-Published) – 543 games; 3,824 publishers
Pig-1: Top 5 Categories	Card Game – 4,615 games
Pig-2: Top 10 Most-Rated	Pandemic – 109,006 users rated
Pig-3: Top 10 Most-Owned	Pandemic – 168,364 owners
Pig-4: Rated by Category	800 output records across 81 categories
Pig-5: Owned by Category	800 output records across 81 categories

7 Conclusion

In this laboratory experiment, a BoardGameGeek dataset was processed using Hadoop MapReduce and Apache Pig. The cleaned dataset contained 21,347 valid game records with 81 primary categories and 3,824 primary publishers.

Three MapReduce programs were used to find the most-owned game, generate rating and engagement summaries, and count games by publisher. Five Pig programs were then used to perform global ranking and category-wise analysis.

Pandemic was the most-owned game with 168,364 owners and also had the highest number of user ratings with 109,006 ratings. Card Game was the largest primary category with 4,615 games. All eight analysis jobs produced valid outputs, and the important results were independently checked using shell commands.

The experiment also demonstrated how Hadoop 3.4.1 and Pig 0.17.0 can be used together on macOS for practical large-dataset processing.

8 GitHub Repository

The complete source code, Pig scripts, result files and screenshots are available at:

https://github.com/tahsin1017/BoardGameGeek-Data-Analysis-using-Hadoop

References

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